SMS Spam Detection

Data Cleaning Report

Report Generated: 2026-07-17 04:39:13

1. Dataset Overview

Total Original Messages: 139,495

Total Cleaned Messages: 9,841

Removed Messages: 129,654 (92.9%)

Label Distribution:

Spam: 1,537 (15.6%)

Ham: 8,304 (84.4%)

Message Statistics:

Average Word Count: 15.62

Average Character Count: 83.55

Maximum Words: 255

Minimum Words: 1

Features Preserved:

Messages with URLs: 234

Messages with Emails: 79

Messages with Phone Numbers: 811

2. Data Cleaning Steps Performed

Step 1: SMS Abbreviation Expansion

Expanded common SMS abbreviations (u->you, pls->please, etc.)

Step 2: Punctuation Spacing Fix

Fixed spacing around punctuation marks for better readability

Step 3: Extra Whitespace Removal

Removed multiple spaces and normalized whitespace

Step 4: Repeated Punctuation Removal

Removed repeated punctuation marks (!!, ??, etc.)

Step 5: Lowercase Conversion

Converted all text to lowercase for consistency

Step 6: Empty Message Removal

Removed messages with no content or very short content (129,654 messages)

Step 7: Duplicate Message Removal

Removed duplicate messages to avoid bias in training

Step 8: URL Preservation

Kept URLs as is without modification (http, https, www)

Step 9: Email Preservation

Kept email addresses as is without modification

Step 10: Phone Number Preservation

Kept phone numbers as is without modification

3. Sample of Cleaned Data

Message 1 - Label: HAM

your opinion about me? 1. over to. jada 3. kusruthi for. lovable 5. silent 6. spl characte...

Message 2 - Label: HAM

what's up? do you want me to come online? if you are free we can talk sometime

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Message 3 - Label: HAM

so you workin overtime nigpun?

Message 4 - Label: HAM

also sir, i sent you an email about how to log into the usc payment portal. i.ll send you ...

Message 5 - Label: SPAM

please stay at home. to encourage the notion of staying at home. all tax-paying citizens a...

Message 6 - Label: SPAM

bankofamerica alert 137943. please follow http://bit.do/cgjk-and re-activate

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Message 7 - Label: HAM

sorry dude. dont know how i forgot. even after dan reminded me. sorry. hope you guys had f...

Message 8 - Label: HAM

i don't quite know what to do. i still can't get hold of anyone. i cud pick you up bout 7....

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4. Output Files Generated

CSV File:

File Name: cleaned_sms_dataset.csv

Columns: 2

- label: Class label (spam/ham)
- cleaned_message: Preprocessed text message

Total Rows: 9,841

File Location: /content/drive/MyDrive/8th sem major project/

PDF Report:

File Name: cleaning_report.pdf

Content: Complete cleaning report with statistics and steps

5. Ready for Machine Learning Training

The cleaned dataset is now ready for machine learning training with 2 columns: label and cleaned_message.

Python Code Example for Model Training:

```
# Load and prepare data
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report

# Load cleaned data
df = pd.read_csv('cleaned_sms_dataset.csv')
X = df['cleaned_message']
y = df['label']

# Vectorize text
vectorizer = TfidfVectorizer(max_features=10000, ngram_range=(1,3))
X_vec = vectorizer.fit_transform(X)

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X_vec, y, test_size=0.2, random_state=42, stratify=y
)

# Train model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Evaluate
y_pred = model.predict(X_test)
print(f'Accuracy: {accuracy_score(y_test, y_pred)*100:.2f}%')
print(classification_report(y_test, y_pred))
```

6. Summary of Cleaning Actions

Action	Details	Count
Original Messages	139,495	
Cleaned Messages	9,841	
Removed Messages	129,654	92.9%
Spam Messages	1,537	15.6%
Ham Messages	8,304	84.4%
URLs Preserved	234	
Emails Preserved	79	
Phones Preserved	811	
Average Words	15.62	
Average Characters	83.55	
Max Words	255	
Min Words	1	

Generated by SMS Detection System - Google Colab